

L11: Deep Learning for Text

CSci 560 Deep Learning w/ Python (Chollet) Ch. 11

Derek Harter

Department of Computer Science
East Texas A&M University

Summer 2025

L11.1 Natural Language Processing: The bird's eye view

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Natural vs. Computer Languages

- In computer science, we refer to human languages, like English or Mandarin, as **natural** languages, to distinguish them from languages designed for machines.
- Every machine language was **designed**: its starting point was a human engineer writing down a set of formal rules to describe the language.
 - Rules came first.
- Natural languages are shaped by an evolutionary process, much like biological organisms, that's what makes them *natural*.
 - Their “rules”, like the grammar of English, were formalized after the fact and are often ignored or broken by its users.

Conclusion: While machine-readable languages are highly structured and rigorous, natural language is messy - ambiguous, chaotic, sprawling, and constantly in flux.

Modern Machine Learning NLP Systems

Using machine learning and large datasets to give computers the ability to *understand* language.

Some examples of types of text processing systems:

- “What’s the topic of this text?” (text classification)
- “Does this text contain abuse?” (content filtering)
- “Does this text sound positive or negative?” (sentiment analysis)
- “What should be the next word in this incomplete sentence?” (language modeling)
- “How would you say this in German?” (translation)
- “How would you summarize this article in one paragraph?” (summarization)

In much the same way that computer vision is pattern recognition applied to pixels, NLP is pattern recognition applied to words, sentences, and paragraphs.

L11.2 Preparing Text Data

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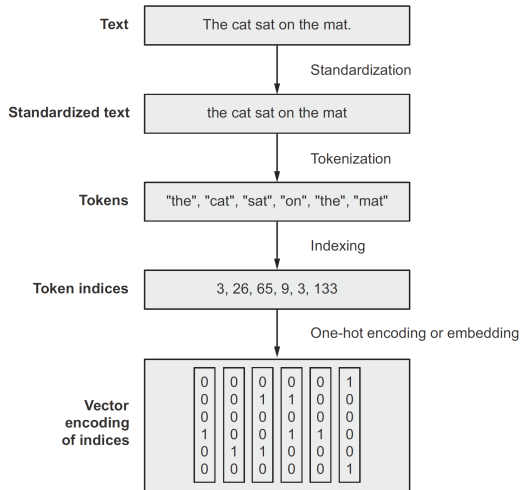
Derek Harter

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Text Vectorization Process

Vectorizing text is the process of transforming text into numeric tensors.



- First, you standardize the text to make it easier to process, such as by converting it to lowercase or removing punctuation.
- You split the text into units (called tokens), such as characters, words, or groups of words. This is called tokenization.
- You convert each such token into a numerical vector. This will usually involve first indexing all tokens present in the data.

Text Standardization

- A machine learning model doesn't know a priori that “i” and “l” are the same letter, that “é” is an “e” with an accent, or that “staring” and “stared” are two forms of the same verb.
- Text standardization is a basic form of feature engineering that aims to erase encoding differences that you don't want your model to have to deal with.
- Simplest and usually used: convert to lowercase and remove punctuation characters.
- Also special characters may be standard form, such as replacing accented vowels with unaccented form.
- **stemming** converting variations of a term into a single shared representation
 - turn “caught” and “been catching” into “[catch]”
 - plurals often stemmed, “cats” into “[cat]”
- With these standardization techniques, your model will require less training data and will generalize better.

Text splitting (tokenization)

Tokenization: break up text into units to be vectorized.

Three different approaches:

- 1 **Word-level tokenization** Where tokens are space-separated (or punctuation separated) substrings. A variant of this is to further split words into subwords when applicable for instance, treating “staring” as “star+ing” or “called” as “call+ed.”
- 2 **N-gram tokenization** Where tokens are groups of N consecutive words. For instance, “the cat” or “he was” would be 2-gram tokens (also called bigrams).
- 3 **Character-level tokenization** Where each character is its own token. In practice, this scheme is rarely used, and you only really see it in specialized contexts, like text generation or speech recognition.

Text-processing Model Types: sequence vs. bag-of-words

There are two kinds of text-processing models:

- ① Those that care about word order are called **sequence models**.
- ② Those that treat input words as a set, discarding original order, are called **bag-of-words models**.

If building a sequence model, you'll use word-level tokenization.

If building a bag-of-words model, you'll use N-gram tokenization.

Understanding N-grams and bag-of-words

- Word N-grams are groups of N (or fewer) consecutive words.
- For example, given sentence “the cat sat on the mat.”
- 2-grams (or bigrams): ['the cat', 'cat sat', 'sat on', 'on the', 'the mat']
- 3-grams (or trigrams): ['the cat sat', 'cat sat on', 'sat on the', 'on the mat']

Because bag-of-words isn't an order-preserving tokenization method (the tokens generated are understood as a set, not a sequence, and the general structure of the sentences is lost), it tends to be used in shallow language-processing models rather than in deep learning models.

Vocabulary Indexing

- Once your text is split into tokens, you need to encode each token into a numerical representation.
- You could potentially do this in a stateless way, such as by hashing each token into a fixed binary vector
- In practice, the way you'd go about it is to build an index of all terms found in the training data
- It's common to restrict the vocabulary to the top N most frequent words, for example 10,000
- When we look up a new token in the vocabulary index, it may not necessarily exist.
 - To handle this, you should use an “out of vocabulary” index (abbreviated as OOV index), a catch-all for any token that wasn't in the index.
 - **OOV token:** “out of vocabulary”, usually token index 1
 - **mask token:** “ignore me not a word”, usually token index 0

L11.3 Two Approaches for Representing Groups of Words: sets and sequences

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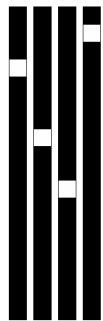
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How to Encode Word Order

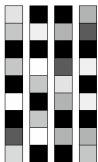
- How to represent **individual words** is relatively uncontroversial: they're categorical features and we assign an integer index (dimension number) and vectorize (encode as dimensions in a feature space).
- More problematic question: how to encode **the way words are woven into sentences**: word order.
- Unlike many other types of timeseries, words in most natural languages don't have a canonical order.
 - Order is clearly important, but its relationship to meaning (in natural languages) isn't straight forward.
- Simplest approach: discard order and treat as unordered set: **bag-of-words models**
- Process strictly in order they appear, like steps in a timeseries: **sequence models**
- Hybrid approach: **Transformer architecture** technically order-agnostic, yet injects word-position info into representations.

Understanding Word Embeddings



One-hot word vectors:

- Sparse
- High-dimensional
- Hardcoded



Word embeddings:

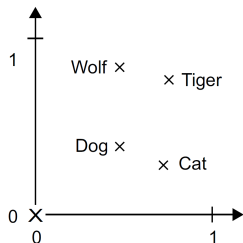
- Dense
- Lower-dimensional
- Learned from data

Figure 2: Word representations obtained from one-hot encoding are sparse, high-dimensional and hardcoded. Word embeddings are dense, relatively low-dimensional, and learned from data. (Chollet,

2021, pg. 330)

- One-hot encoding assumes that different tokens are **all independent from each other**. (one-hot vectors are all orthogonal to one another).
- The **geometric relationship** between two word vectors should reflect the **semantic relationship** between these words.
 - In a reasonable word vector space, similar words should be encoded in similar vectors.
 - e.g. the vector for “film” and “movie” should be about the same vector.
- **Word embeddings** are vector representations of words that achieve exactly this: they map human language into a structured geometric space.

Example of Word Embedding Space



- 4 words embedded on a 2D plane: cat, dog, wolf and tiger
- Same vector from “dog” to “wolf” and “cat” to “tiger”
 - semantics “from pet to wild animal”
- Similar same vector from “dog” to “cat” as “wolf” to “tiger”
 - semantics “from canine to feline”

Figure 3: A toy example of a word embedding space. (Chollet, [2021](#), pg.330)

Two Ways to Obtain Word Embeddings

- Learn word embeddings jointly with the main task you care about (such as document classification or sentiment prediction).
 - In this setup, you start with random word vectors and then learn word vectors in the same way you learn the weights of a neural network.
- Load into your model word embeddings that were precomputed using a different machine learning task than the one you're trying to solve.
 - These are called pretrained word embeddings.

Summary of Representing Groups of Words

- Word order in Text processing is handled in 2 basic ways:
 - ① **bag-of-words models** : discard order and treat as an unordered set (multi-hot encoding typically, 20,000 sparse vectors).
 - ② **sequence models**: process words in order they appear like a timeseries
- For sequence models, can encode sequence again using one-hot encoding, but this ends up with very large input to RNN models.
- **word embeddings** are vector representations of words that map humanlanguage into a structured geometric space.



L11.4 The Transformer Architecture

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L11.5 Beyond Text Classification: Sequence-to-sequence learning

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Chollet, F. (2021). *Deep learning with python* (second). Manning.